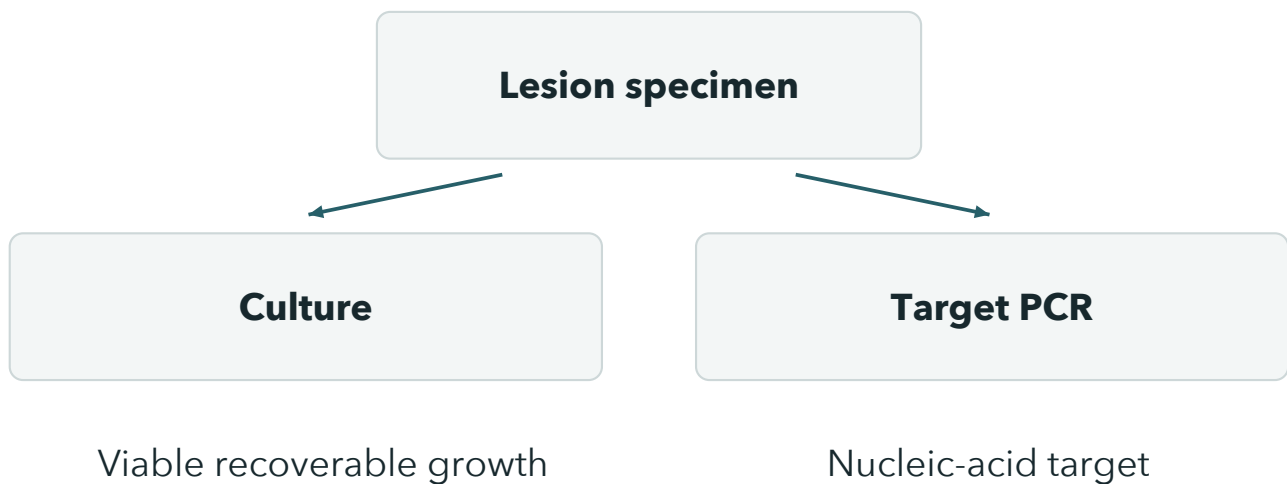


Microbiological tests and specimen logic

Part 1 | DNA detection and viable growth differ

A molecular result and a culture result can diverge without either being intrinsically contradictory.



Culture needs viable recoverable organisms; PCR detects its nucleic-acid target under the assay conditions.

After antimicrobial exposure

Residual target DNA can remain detectable when viable recoverable organisms have fallen. Conversely, inadequate sampling, target mismatch or inhibition can produce molecular negatives. Interpret discordance with treatment timing and specimen quality instead of treating PCR as a direct count of living pathogens.

Sample the biological compartment

A test cannot identify an organism that never reaches the specimen.

Limitation

Feature	Problem	Targeted response
Superficial specimen	Deep lesion not sampled	Specialist depth-targeted sampling
Very small volume	Low organism yield	Plan specimen allocation
Prior treatment	Reduced viable growth	Document therapy and timing

Reasoning sequence

First ask whether the correct tissue and depth were sampled. Then assess transport and assay suitability. Repeating an identical poor specimen can give a reproducibly negative result without providing strong evidence that the lesion is noninfectious.

A test result needs organism context

Different methods detect different structures and growth requirements.

Method

Feature	What it contributes	Limit
Microscopy	Morphology and staining	Sensitivity depends on burden
Culture	Recoverable organism and susceptibility	Growth conditions matter
PCR	Targeted nucleic acid	Target and inhibition matter

Compare complementary evidence

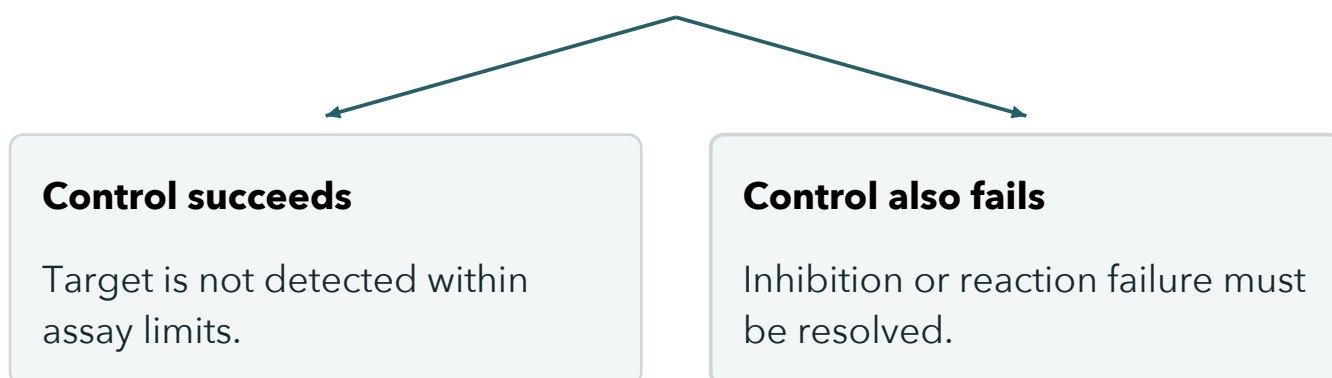
Morphology can guide which assays or media to request, while culture can answer a susceptibility question that a simple species-target PCR does not. A molecular panel's negative result is not an exclusion of every possible organism outside its target list.

An internal control tests the reaction

Target absence and reaction failure have different interpretations.

Target PCR is negative

Did the internal control amplify?



What the control cannot guarantee

A post-extraction amplification control can reveal reaction inhibition but does not prove that the lesion was sampled adequately. An extraction-process control can assess additional steps. Identify when the control entered the workflow before deciding which failure it can exclude.

Design a repeat that adds information

A changed specimen or corrected inhibition is more informative than repetition alone.

Interpret a discordant result

- Review phenotype and treatment
↓
- Check specimen site, depth and handling
↓
- Inspect controls and target coverage
↓
- Choose a repeat that addresses the identified limitation

An original comparison

One repeat uses another superficial aliquot with the same failed internal control; another resolves inhibition and samples the actual lesion. Only the second addresses the demonstrated limitations. The lesson is to test a reason for the first negative, not merely to accumulate negative reports.